PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Akanbi Temitope Benjamen
Program/Batch	B084-Networkwalks
Date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

All activities documented in this report were conducted strictly for educational and research purposes on personally owned systems or systems for which written authorization was obtained. The findings are intended solely for learning and defensive security awareness and must not be used for unauthorized or unlawful activities. The individual conducting the tests is fully responsible for ensuring compliance with applicable laws. Neither the instructor, authors, nor Networkwalks shall be held responsible for any misuse of the information provided.

2. Introduction

This report presents the results of the Week 2 footprinting and reconnaissance practical completed as part of the Networkwalks Cybersecurity Programme, Batch B082. The exercise focused on gathering publicly observable information about the networkwalks.com domain by using several command-line tools available in Kali Linux.
The practical involved WHOIS, WhatWeb, nslookup, curl, Wafw00f and DNSRecon. Each utility was used to examine a different aspect of the target, including domain registration details, website technologies, DNS resolution, HTTP response headers, web application firewall identification and DNS records.
The sections that follow describe the commands executed, the results returned and the relevance of each observation from a cybersecurity perspective. The findings are based on the terminal output captured in the submitted results document.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I conducted footprinting and reconnaissance on the **networkwalks.com** domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon**. Each tool was used to gather specific information about the target domain and its web infrastructure.

I began with **WHOIS** to retrieve publicly available domain registration details, including the registrar, registration dates, and name servers.

Next, I used **WhatWeb** to identify the technologies powering the website. The results revealed that the website uses **WordPress 7.1.4** and **WordPress Download Manager 3.3.76**, alongside other web technologies.

With **Nslookup**, I queried the domain's DNS information and successfully resolved networkwalks.com to the IP address **192.232.216.135**.

I then used **Curl** with the -I option to examine the website's HTTP response headers. The output provided useful information about the web server and revealed the presence of the WordPress REST API endpoint (/wp-json/).

Afterwards, I used **Wafw00f** to check for Web Application Firewall protection. The result indicated that the website is protected by **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate the domain's DNS records. This revealed information about its name servers, mail server, SPF and TXT records, service records, and DNS configuration.

4.2 Network Scanning with Zenmap

For the second activity, the practical involved using **Zenmap** to discover active devices on a local network and examine their network information.

The process involved using the Windows ipconfig command to identify the local IP address and subnet. The identified subnet was then entered into Zenmap, where the **Ping Scan** option was selected to discover active hosts.

The practical's sample output displayed four live hosts:

- 10.0.0.1
- 10.0.0.4
- 10.0.0.19
- 10.0.0.5

The sample also displayed the corresponding MAC address information for the discovered devices.

Finally, the practical involved opening the **Topology** section in Zenmap, enabling the legend, and exporting the generated network topology as a PDF document.

Note: The IP addresses and host details listed above are sample results from the practical instructions. They should not be presented as actual findings unless they match the results obtained from my own network scan.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Website technology information exposed	WhatWeb identified WordPress and WordPress Download Manager.	Exposed technology details could help attackers identify software versions that may require further security assessment.	Medium
2	Publicly identifiable server IP address	Nslookup resolved networkwalks.com to 192.232.216.135.	This information reveals the IP address associated with the website's hosting infrastructure.	Low
3	Web server information exposed	Curl retrieved HTTP response headers and identified the /wp-json/ endpoint.	The exposed information may support technology identification and further reconnaissance.	Low
4	Web Application Firewall identified	Wafw00f detected ModSecurity (SpiderLabs).	Disclosure of the WAF technology provides insight into the website's security infrastructure.	Low
5	DNS infrastructure details exposed	DNSRecon identified name server, mail server, SPF/TXT, and service records.	The information could help build a more detailed picture of the domain's network infrastructure.	Medium
6	Multiple active devices detected on local network	The practical's sample Zenmap results displayed four live hosts.	Unidentified devices on a network may introduce potential security concerns if they are not properly monitored.	Medium

Risk Level Key: Critical | Medium | Low

The findings listed above represent observations made during the reconnaissance and scanning exercises; they should not be interpreted as confirmed vulnerabilities.

The activities focused primarily on information gathering and network host discovery. No exploitation or vulnerability validation was conducted during these exercises.

Therefore, the identification of software versions, IP addresses, DNS records, or active network devices does not automatically indicate a security weakness. Further authorized assessment would be necessary to establish whether any of these observations present an actual security risk.

6. Recommendations

Based on the findings from the footprinting and network scanning activities, the following security measures are recommended to improve the protection of web applications and network infrastructure.

1. **Review Publicly Exposed Technology Information**
   Organizations should periodically assess the technical information exposed by their websites, including CMS platforms, plugins, and other web technologies.
2. **Maintain Updated Software**
   CMS platforms, plugins, and other web components should be updated regularly to address known security issues and maintain system integrity.
3. **Strengthen HTTP Header Configuration**
   Web server response headers should be reviewed and properly configured to minimize unnecessary disclosure of technical information.
4. **Conduct Regular DNS Record Reviews**
   DNS records should be examined periodically to ensure that publicly accessible information and service configurations remain appropriate.
5. **Maintain Proper WAF Configuration**
   The existing Web Application Firewall (ModSecurity) should remain enabled, regularly maintained, and properly configured to support web application security.
6. **Conduct Periodic Network Discovery**
   Organizations should perform authorized network scans regularly to identify active devices and maintain visibility across their networks.
7. **Verify Unknown Network Devices**
   Any unfamiliar or unexpected devices discovered during network scans should be investigated to confirm their identity and authorization.
8. **Keep Network Documentation Updated**
   Network topology, device details, and relevant infrastructure information should be properly documented and reviewed regularly.
9. **Ensure Proper Authorization for Security Testing**
   All reconnaissance and scanning activities should be conducted only on personally owned systems or networks for which appropriate permission has been granted.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how **WHOIS** can provide domain information, **WhatWeb** can identify web technologies, **Nslookup** can resolve domain names, **Curl** can inspect HTTP headers, **Wafw00f** can identify a WAF, and **DNSRecon** can provide additional DNS information.

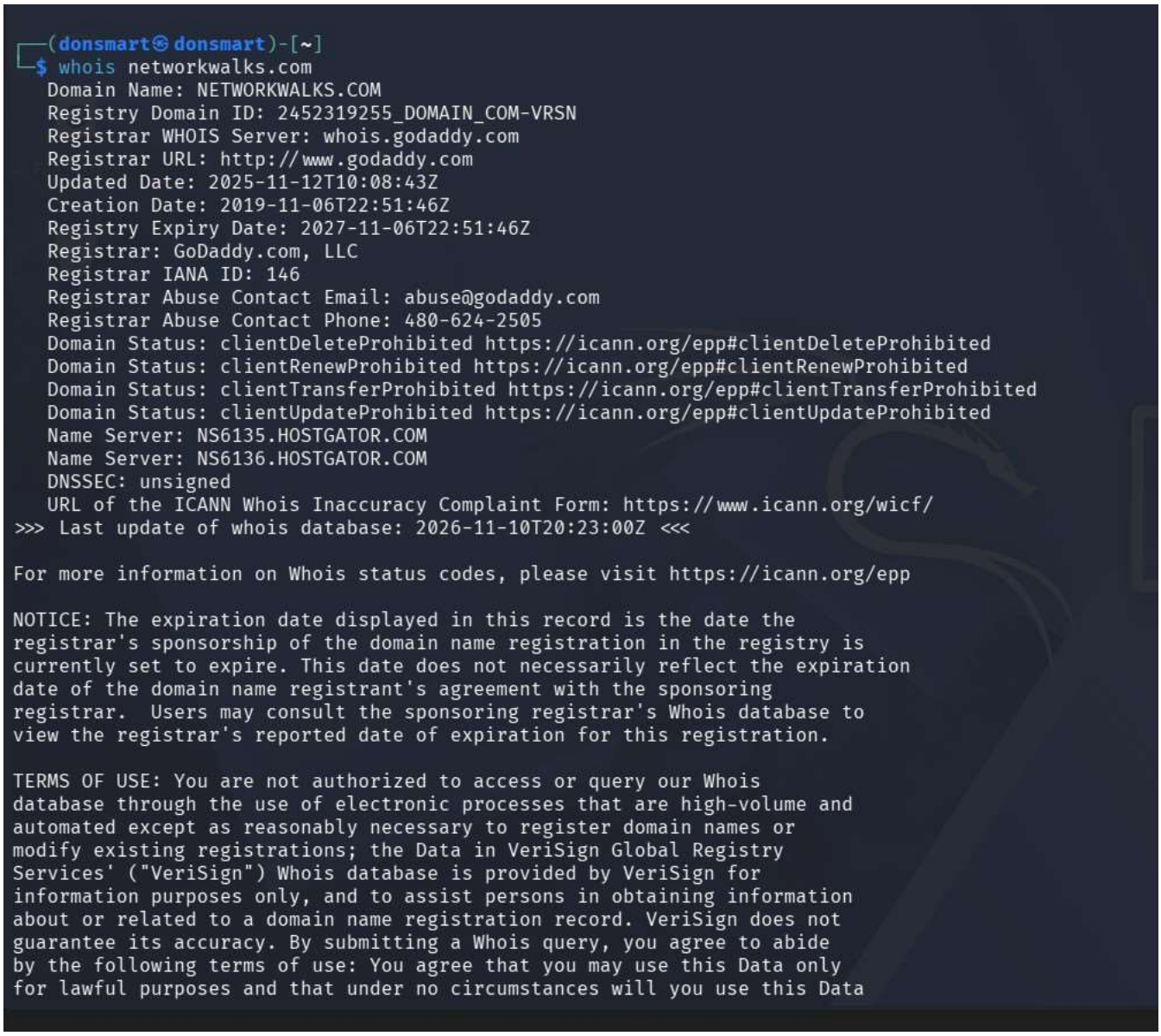
In the network scanning activity, I used **Zenmap** to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

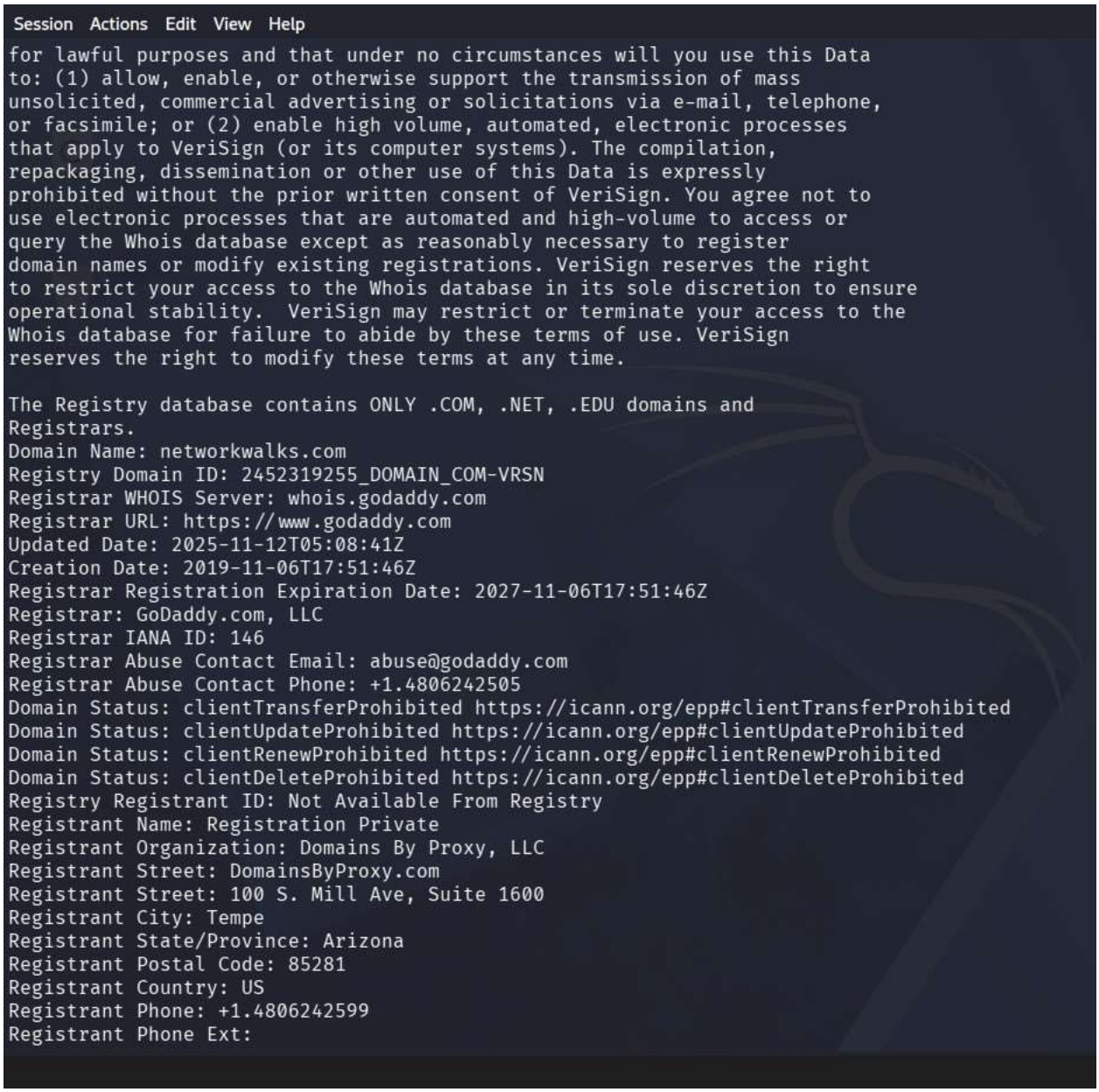
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected




Session Actions Edit View Help
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:

```
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM
Name Server: NS30.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-11-10T20:23:13Z <<<
For more information on Whois status codes, please visit https://icann.org/epp

TERMS OF USE: The data contained in this registrar's Whois database, while believed by the
registrar to be reliable, is provided "as is" with no guarantee or warranties regarding its
accuracy. This information is provided for the sole purpose of assisting you in obtaining
information about domain name registration records. Any use of this data for any other purpose
is expressly forbidden without the prior written permission of this registrar. By submitting
an inquiry, you agree to these terms and limitations of warranty. In particular, you agree not
to use this data to allow, enable, or otherwise support the dissemination or collection of this
data, in part or in its entirety, for any purpose, such as transmission by e-mail, telephone,
postal mail, facsimile or other means of mass unsolicited, commercial advertising or solicitations
of any kind, including spam. You further agree not to use this data to enable high volume, automated
or robotic electronic processes designed to collect or compile this data for any purpose, including
mining this data for your own personal or commercial purposes. Failure to comply with these terms
may result in termination of access to the Whois database. These terms may be subject to modification
at any time without notice.

**NOTICE** This WHOIS server is being retired. Please use our RDAP service instead.
```

```
┌──(donsmart㉿donsmart)-[~]
└─$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
```

```
┌──(donsmart㉿donsmart)-[~]
└─$ nslookup networkwalks.com
Server:         10.162.176.213
Address:        10.162.176.213#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135


┌──(donsmart㉿donsmart)-[~]
└─$
```

```
┌──(donsmart㉿donsmart)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https:
//recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "h
ttps://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https
://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/
53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=6d5ce715ce676e511abfbc4f46629b02; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 21:06:56 GMT
server: Apache

┌──(donsmart㉿donsmart)-[~]
└─$
```


```
┌──(donsmart㉿donsmart)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https:
//recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "h
ttps://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https
://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/
53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=6d5ce715ce676e511abfbc4f46629b02; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Tue, 10 Nov 2026 21:06:56 GMT
server: Apache

┌──(donsmart㉿donsmart)-[~]
└─$
```


```
┌──(donsmart㉿donsmart)-[~]
└─$ wafw00f networkwalks.com

                 _____
                /     \
               (  Woof! )
                \  ___/
                ,,
            .-. -    _____
           ()``; |==|_____)
           / ('       /|\
      (  /  )        / | \
       \(_)_))      /  |  \
                                    )
                                   ) (_
                                  ( |__|
                                .)|__|
                               (  |__|
                                . |__|
                                  |__|

                ~ WAFW00F : v2.4.2 ~
    The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

┌──(donsmart㉿donsmart)-[~]
└─$
```

```
┌──(donsmart㉿donsmart)-[~]
└─$ dnsrecon -d networkwalks.com
2026-11-10T22:16:02.485043+0100 INFO Starting enumeration for domain: network
walks.com
2026-11-10T22:16:02.485702+0100 INFO std: Performing General Enumeration agai
nst: networkwalks.com...
2026-11-10T22:16:03.250891+0100 ERROR No answer for DNSSEC query for networkw
alks.com
2026-11-10T22:16:04.374605+0100 INFO     SOA ns6135.hostgator.com 50.87.144.8
7
2026-11-10T22:16:05.292092+0100 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-10T22:16:05.916082+0100 INFO     Bind Version for 50.87.144.87 "9.16.
23-RH"
2026-11-10T22:16:05.917122+0100 INFO     NS ns6136.hostgator.com 192.232.216.
131
2026-11-10T22:16:06.622515+0100 INFO     Bind Version for 192.232.216.131 "9.
16.23-RH"
2026-11-10T22:16:07.852926+0100 INFO     MX mail.networkwalks.com 192.232.216
.135
2026-11-10T22:16:07.862296+0100 INFO     A networkwalks.com 192.232.216.135
2026-11-10T22:16:08.856315+0100 INFO     TXT networkwalks.com v=spf1 +a +mx +
ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-10T22:16:08.856638+0100 INFO     TXT networkwalks.com google-site-ver
ification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-10T22:16:09.721876+0100 INFO Enumerating SRV Records
2026-11-10T22:16:17.660053+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-10T22:16:17.661222+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-10T22:16:17.661460+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-10T22:16:17.661609+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-10T22:16:17.661796+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-10T22:16:17.661953+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-10T22:16:17.663128+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-10T22:16:17.663468+0100 INFO     SRV _autodiscover._tcp.networkwalks.
com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-10T22:16:17.664901+0100 INFO 8 Records Found
2026-11-10T22:16:17.665416+0100 INFO Completed enumeration for domain: networ
kwalks.com


┌──(donsmart㉿donsmart)-[~]
└─$
```

Zenmap

Scan Tools Profile Help

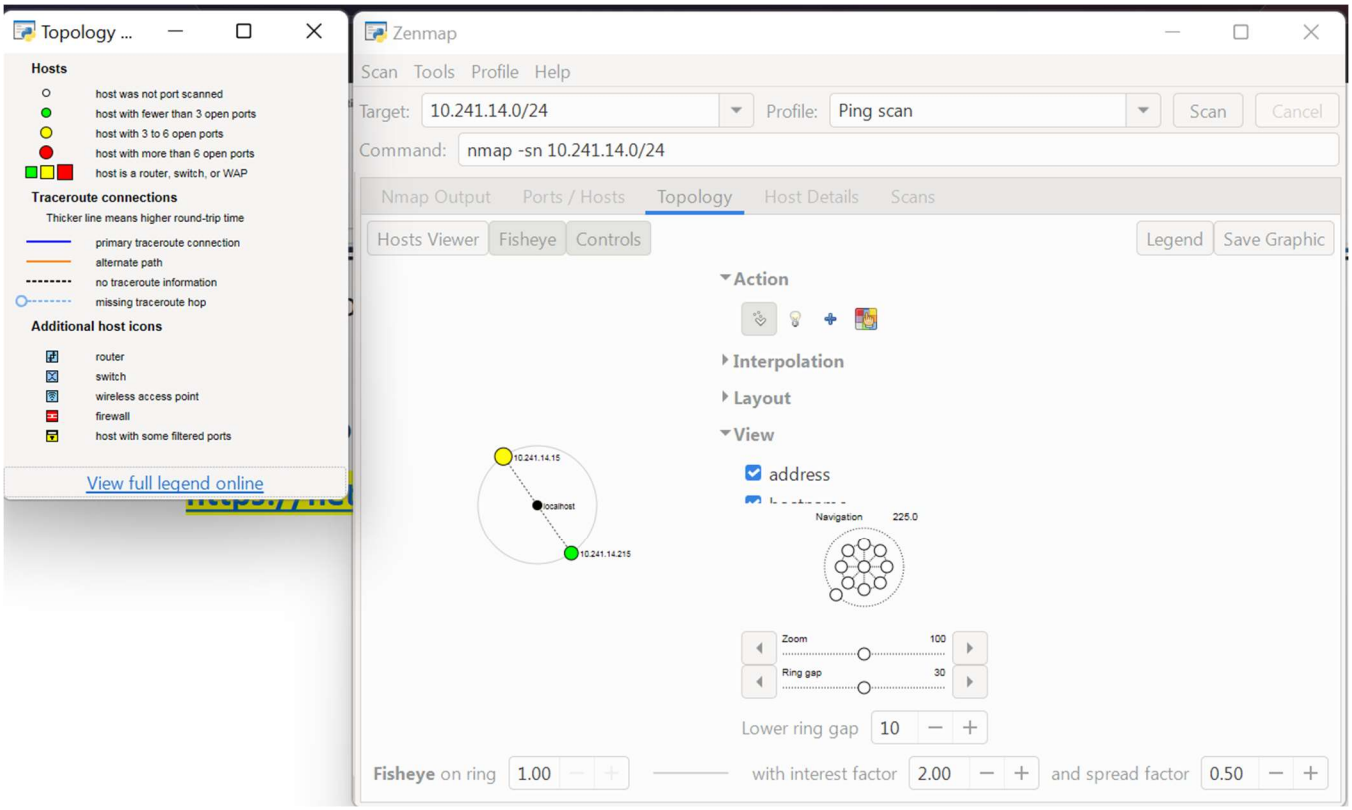
Target: 10.241.14.0/24 Profile: Ping scan Scan Cancel

Command: nmap -sn 10.241.14.0/24

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -sn 10.241.14.0/24 Details

```
Starting Nmap 7.993 ( https://nmap.org ) at 2026-11-11 22:52 +0100
Nmap scan report for 10.241.14.215
Host is up (0.0040s latency).
MAC Address: 9E:78:84:9E:A4:E0 (Unknown)
Nmap scan report for 10.241.14.15
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 8.11 seconds
```



-End-

👤 Author

Akanbi Temitope Benjamen
Cybersecurity Professional B084
LinkedIn: https://www.linkedin.com/in/temitope-akanbi-749363442/

📌 Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub